Task: Using statistical analysis, determine whether there is a causal relationship between ice cream sales and shark attacks

Visualizing the relationships

Day	Temperature (°C)	Shark Attacks	Ice Cream Sales
1	28	1	120
2	30	2	150
3	32	3	180
4	29	1	130
5	33	4	200
6	27	0	110
7	31	2	170
8	34	5	210
9	26	0	100
10	35	6	220

Visualization 1:

Temperature	Sum of Ice Cream Sales
26	100
27	110
28	120
29	130
30	150
31	170
32	180
33	200
34	210
35	220
Grand Total	**1590**

What do we have here?

It looks like that the higher the temperature gets – the more ice-cream sales are.

For example: At 26 degrees – there are 100 ice-cream sales. At 27 degrees – 110. (with an additional 10)

At 35 degrees – 220 sales.

Conclusion: The people are looking more for ice-cream, when it's hotter.

Visualization 2:

Temperature	Sum of Shark Attacks
26	0
27	0
28	1
29	1
30	2
31	2
32	3
33	4
34	5
35	6
Grand Total	**24**

The situation here:

The higher the temperature gets – the more shark attack are there.

For example: at 26 degrees – no shark attacks. But at 28 – 1 attack. At 35 degrees – 6 attacks.

Conclusion: The shark attack more with the increasing of the temperature.

People – Beware! And be careful!

Visualization 3:

Shark Attacks	Sum of Ice Cream Sales
0	210
1	250
2	320
3	180
4	200
5	210
6	220
Grand Total	**1590**

And here:

It looks like that there is no connection between the shark attack and the number of the ice-cream sales. But they both depend on the temperature!

1 person can by 10 ice-creams if he wants or even 20.

The important is that there are possible shark attacks during the summer, so watch out!

So, what is the conclusion?

We no know that the higher the temperature gets – there are the more ice-cream sales. Also the higher the temperature gets – more shark attacks are possible. But the sales do not depend on the number of the attack.

Correlation Matrix

	Temperature	Ice Cream Sales	Shark Attacks
Temperature	1.00	1.00	0.98
Ice Cream Sales	1.00	1.00	0.97
Shark Attacks	0.98	0.97	1.00

This is the correlation matrix, generated with the help of AI:

Temperature <> Ice Cream Sales – it’s a perfect correlation – The higher the temperature gets, the more sales of ice-cream are.

Temperature <> Shark Attacks – it’s almost perfect correlation – The higher is the temperature, there more shark attack are possible.

Ice Cream Sales <> Shark Attacks – it says 0.97 – almost perfect, but it is doubtful. Both depend on the temperature. And that is the confusion variable here – the temperature.

Here is a Python programming code for forming the Correlation Matrix

```
import pandas as pd

# Table Data
data = {
   'Temperature': [28, 30, 32, 29, 33, 27, 31, 34, 26, 35],
   'Ice Cream Sales': [120, 150, 180, 130, 200, 110, 170, 210, 100, 220],
   'Shark Attacks': [1, 2, 3, 1, 4, 0, 2, 5, 0, 6]
}

# Creating of DataFrame
df = pd.DataFrame(data)

# Calculating the correlation matrix
correlation_matrix = df.corr()

# Showing the result
print("📊 Correlation Matrix:")
print(correlation_matrix.round(2))
```

Ice cream sales *cause* shark attacks – or there is something else?

Shark Attacks	Sum of Ice Cream Sales
0	210
1	250
2	320
3	180
4	200
5	210
6	220
Grand Total	**1590**

We are going back to Visualization 3. We clearly see that there is no permanent increase of the ice-cream sales with the shark attacks, but……….

What can we see? There is sometimes increasing of the sales with the shark attack quantity increasing. It is very likely sometimes the number of the attacks to depend on the number of sales. Still, we already know that both depend on the temperature.

It is really possible the attacks to be connected to the sales. There is one big question – Are the sharks periodically going back to the waters for some reason? For different environment? Because they are getting bored if the stay near the shore with less water? Or perhaps the more quantity of water is there, the better they breathe?

Perhaps. A sea biologist rather will give an answer to these questions. Such specialist could tell us. But by that time, it is so much important to be very careful and to be informed. Unfortunately today there are negative news, related to shark attacks and even deaths. That should give us a good enough lesson not to take risks, otherwise some day a dark moment may come for us. It is important to know what we do, because if we don't – that could cost us even our whole life.

Be informed, be educated, be an example to all people around you!

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During the analysis the following platforms and programs were used:

- Microsoft Word;
- Microsoft Excel;
- Python;
- My SQL;
- AI (for explaining and better understanding the case)